

Phase 2: Requirement Analysis

OptiCrop: Smart Agricultural Production Optimization

Role	Name
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Software Dependencies

- **Operating System:** Windows 10/11, macOS, or Ubuntu Linux
- **Runtime Environment:** Python 3.8 or higher
- **Package Manager:** pip (Python package installer)
- **External Packages:** Flask 3.0.0, Scikit-Learn 1.3.2, Pandas 2.1.4, NumPy 1.26.2, Matplotlib 3.8.2, Seaborn 0.13.0, Jupyter Notebook 5.0.0

Hardware Configurations

- **Processor:** Any dual-core processor (Intel i3/i5 or AMD equivalent)
- **RAM:** Minimum 4 GB RAM (8 GB recommended for smooth model training)
- **Storage:** Minimum 500 MB free disk space for datasets, Python environments, and serialized models (.pkl files)

Data Specifications

The system utilizes the Crop Recommendation dataset.

- **Format:** Comma Separated Values (CSV)
- **Total Records:** Approximately 2,200 rows

- **Features (7 Independent Variables):**
 - Nitrogen (N) - Integer/Ratio
 - Phosphorous (P) - Integer/Ratio
 - Potassium (K) - Integer/Ratio
 - Temperature - Float/Continuous
 - Humidity - Float/Continuous
 - pH - Float/Continuous
 - Rainfall - Float/Continuous
- **Target Variable (1 Dependent):** Label (Categorical - String representing 22 different crop types)